

ORIGINAL ARTICLE

Computerized training of working memory in a group of patients suffering from acquired brain injury

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Abstract

Objective: Study short- and long-term transfer effects of a computerized working memory (WM) training programme for patients suffering from working memory deficits after acquired brain injury.

Research design: A controlled experimental study with a cross-over design.

Methods: A consecutive sample of 21 subjects. Mean age 43.2 years, time since injury/illness onset 37 months (median). The subjects were randomly selected into two groups where one group served as controls. All subjects trained daily for 5 weeks in a computer WM task program. They were followed-up at 4 and 20 weeks after the training.

Results: A significant improvement in the trained WM tasks, significant improvements in neuropsychological WM-test results at 4 and 20 weeks after training compared to baseline. Improvement in the subjects' rated occupational performance and satisfaction with performance in pre-defined occupational problems. Rated quality-of-life did not change. However, rated overall health increased 20 weeks after training.

Conclusions: Structured and intense computerized WM training improves subjects' cognitive functioning as measured by neuropsychological WM-demanding tests, rated occupational performance, satisfaction with performance and rated overall health. The training probably has an impact on the rehabilitation outcome, returning to work, as well as on daily activities for individuals with verified WM impairments.

Keywords: Working memory, acquired brain injury, neuropsychology, cognitive rehabilitation, computerized training, occupational performance, QoL

Introduction

Working memory (WM) is the cognitive ability to retain limited information for a short time while one has to manipulate its use [1–5]. WM is closely associated with attention and is needed in most mental activities. It activates relevant memory content such as experience and knowledge and integrates it with current information and then initiates an action. WM is required for updating information and making use of it for thinking and problem-solving as well as for listening in acoustically demanding

situations [6, 7]. A distinction can be made between simple and complex WM tasks. Simple tasks like the Digit span test [8] emphasize storage functions and serial order at retrieval, whereas complex tasks like the Listening span tasks in addition require semantic processing of the items to-be-recalled [6, 7]. Complex WM tasks demand more executive functions because they involve shifting of attention between for instance rehearsal processes and semantic judgements and they presumably tap into multi-tasking relevant for everyday activities.

Frontal and parietal cortices are important for WM function and changes in cortical activity can be seen after WM training [9, 10]. Olesen et al. [11] found that brain activity increased in medial frontal and parietal cortical regions in healthy young people after 5 weeks training on the computerized program. Recently, McNab et al. [12] found cortical changes in prefrontal and parietal dopamine receptors after 5 weeks of WM training. The same cortical regions are also considered important for WM function [13–15]. WM deficits may occur after different diseases and medical conditions affecting certain regions in the brain.

Cognitive processing, including reasoning and problem-solving, requires continuous information processing and integration of information with previous knowledge and skills. To perform an action adequately, one has to prioritize one action among several alternatives. In addition, continuous attention is needed to be sure that the action has attained the intended goal and one has to evaluate the cognitive processing outcome. WM is an essential part of this process. Thus, many daily activities depend on the individual's WM functioning. Rehabilitation outcome and prognosis for returning to work after acquired brain injury also depend on the individual's WM functioning [16–19].

Working memory impairment is one of the most common symptoms after acquired brain injury [19–21]. Patients suffering from mild acquired brain injury, thus patients who can 'walk and talk', often suffer from an impaired WM that is manifested as impaired divided and focused attention, impaired flexibility to change strategy and difficulties choosing between competing alternatives. The patients report the difficulties as impaired concentration, 'losing the thread' and 'not being good at juggling many balls'. The problems affect returning to work, study results and daily functioning.

In a recently published review of multi-disciplinary rehabilitation from acquired brain injury in adults of working age [22], the authors state that little evidence has been presented regarding the effectiveness of multi-disciplinary rehabilitation [22]. It is often hard to draw general conclusions from intervention outcomes since brain injured people have a variety of difficulties with a variability also in severity of the injury. Cognitive rehabilitation is an individualized and long-term process. The authors suggest that further studies be conducted in the context of routine clinical practice [23]. The contents and settings in outpatient units differ, which makes a comparison of cognitive rehabilitation outcomes difficult. Specific training in computer-based WM training is a fairly new type of intervention and there is a lack of published research in this area. However, in a recently published

double-blind controlled study, Flavia et al. [24] showed that the training group improved significantly in neuropsychological tests including working memory tests, while the controls did not.

Usually, patients with memory impairments are trained to use compensatory strategies and to handle memory aids [25–27]. However, WM cannot be trained using conventional memory training [28]. From a clinical perspective and from a theoretical point of view it is important to study whether it is possible to train WM in patients whose WM impairment is pronounced and one of a few remaining symptoms after brain injury. Working memory training focuses on attention-demanding tasks that activate neural structures, which are important for WM function.

The patients who participated in the present study suffered from acquired brain injury. They were enrolled in the cognitive rehabilitation programmes at the Department of Rehabilitation Medicine, University Hospital, Linköping, Sweden. The purpose of these rehabilitation programmes, which were not computerized training, was to enhance the patients' knowledge and awareness about their capacities and impairments and to learn compensatory techniques that help them handle their cognitive problems and fatigue. Nevertheless, after completing the rehabilitation programmes many patients still struggle with WM deficits, causing problems in WM-demanding social situations, in daily problem-solving situations such as understanding and carrying out complex instructions and searching information on the Internet. Many patients are motivated to improve their WM capacity and they ask for training.

A computerized training program (Robomemo) has been used to train WM in children and young people [29, 30]. Intensive and structured training not only improved the results of trained tasks, but importantly also the results for non-trained WM demanding tasks such as attention, problem-solving and also on motor control and behaviour [30–32]. The effect lasted for at least 3 months. Holmes et al. [33] recently found that WM training increased mathematic and learning capacity in school children and the effect lasted for at least 6 months. The training effect depends on training structure and intensity and the training is individually and continuously adapted to the progress and cognitive capacities of each participant.

The training program, QM (formerly called ReMemo), for adults was developed at the Karolinska Institute and Cogmed Cognitive Medical System AB, Sweden [34]. Memory stimuli are presented in a computerized system with visuo-spatial and verbal WM tasks [34].

Verbal and visuo-spatial WM demanding tests were used in previous studies [30, 35].

However, there are tests for even more combined WM functions to be used for studying whether WM training can transfer to WM tasks that are more complex in terms of storage and processing. Complex WM tasks are more important for problem-solving and communicative activities in daily life situations. In addition, whether WM training has a long-term effect is a question that needs further exploration.

One purpose of the present study was to examine short- and long-term effects of a computerized WM training programme for adult patients suffering from WM impairments after acquired brain injury. A second purpose was to study whether WM training can be transferred to complex WM tasks, including dual demands on storage and processing capacities, to WM-related activities and to patients' ratings of their quality-of-life and health.

Method

Subjects

A consecutive sample of patients was selected from 76 persons with acquired brain injury, not in progression, who received outpatient rehabilitation services at the Department of Rehabilitation Medicine, University Hospital in Linköping, Sweden. Criteria for inclusion were as follows: age 20–65 years, passed post-acute state (time since injury/illness ≥ 1 year), self-reported WM impairments and a significantly impaired WM index compared to the index for verbal comprehension and/or index for perceptual organization or a WM index < 80 using the WAIS III [8] as well as a reported motivation for training. Exclusion criteria were $IQ \leq 70$ based on WAIS III, depression according to DSM-IV [36] and perceptual or motor impairments that made computerized training impossible. Thirty-nine subjects fulfilled the inclusion criteria. Two patients were excluded due to extensive cognitive impairments and two were excluded because they had not completed other rehabilitation programmes. Seven individuals declined to participate due to work and six declined due to the long distance from their home (they thought they would be too tired to travel to the training) and one declined due to economic problems. He said he could not afford the travelling costs.

The ability to benefit from cognitive training may vary depending on what kind of neuropsychological deficits the individual suffers from. However, this study focused on WM impairment in a heterogeneous group with different injury-related diagnoses. Twenty-one individuals accepted and participated in the study. As seen in Tables I and II patients were diagnosed with traumatic brain injury, stroke

Table I. Diagnoses for 21 patients, 10 men (48%) and 11 women (52%).

Diagnosis	<i>n</i>	%
Trauma	1	4
Stroke	11	52
Infection (encephalitis)	5	24
Tumour	2	10
Subarachnoidal haemorrhage (SAH)	2	10
Total	21	100

Table II. WM training results for 21 individuals. Start index (baseline), Max index and Improvement.

Individuals	Start index	Max index	Improvement index	Diagnosis
1	57	74	17	Tumour
2	64	87	23	Stroke
3	65	77	12	Stroke
4	65	85	20	Stroke
5	66	82	15	Stroke
6	66	91	25	SAH
7	69	85	16	Trauma
8	69	78	9	Stroke
9	69	84	15	Stroke
10	70	88	18	Infection
11	71	91	20	Infection
12	71	91	20	Stroke
13	72	98	26	Infection
14	77	103	26	Stroke
15	79	94	15	Stroke
16	80	96	15	Infection
17	81	95	14	Tumour
18	84	99	15	SAH
19	85	120	35	Stroke
20	93	123	30	Stroke
21	94	120	26	Infection
<i>M</i>	73.7	93.4		
<i>SD</i>	9.7	13.7		

Table III. Demographic data for 21 patients, 10 men (48%) and 11 women (52%).

	<i>M</i>	<i>SD</i>
Age (27–63)	43.3	9.8
Education	12.7	2.0
Months since illness/injury*	46.4	31.9

*Median 37 months.

(tromboembolic infarction or intracerebral haemorrhage), infection (encephalitis) tumour and subarachnoidal haemorrhage caused by an aneurysm rupture (SAH). Demographic data are presented in Table III. Mean age was 43.2 years (range 27–63)

and time since injury/illness onset was 37 months (median). Four individuals (19%) were working or at preparatory work during training. Index for WAIS III verbal comprehension was $M=96.9$ ($SD=11.8$); index for WAIS III perceptual organization was $M=96$ ($SD=14.9$) and index for WAIS III WM was $M=89.7$ ($SD=8.1$). Verbal comprehension index and perceptual organization index differed significantly from WM index, $p=0.01$ and $p=0.04$, respectively. Sixteen subjects had participated in rehabilitation group programmes focusing on awareness and coping strategies. No one had any experience from cognitive training similar to the current computerized WM training.

Procedure

Questionnaire. The individuals answered a questionnaire that focused on indications for WM training before starting the training sessions. The questionnaire included questions about WM function in daily activities [37].

WM tests. A neuropsychologist assessed all subjects before training (baseline) and at 4 and 20 weeks after training. The assessment used well-known neuropsychological tests that focused on verbal and visual WM.

- *Paced Auditory Serial Attention Test (PASAT 2.4)* [38] tests divided attention, storage of memory elements, mental transformations and responding.
- *Colour Word Interference Test condition 4-Inhibition/Switching (CWIT)* [39] is a Stroop test containing both an interference condition [40] and a switching procedure. It taps both verbal inhibition and executive switching.
- *Block-Span-board forwards and backwards from WAIS R-NI* [41, 42] contains 10 blocks that are placed irregularly on a board. The test leader points at the blocks in a certain sequence and the subject repeats the sequence by pointing in the same order (forwards) or in the reverse order (backwards). Span lengths (the sequence of blocks) are two-to-eight blocks long.

In addition, WM tests requiring increased cognitive load and task complexity were used rather than tests used in previous studies on WM training. Both tasks are dual tasks in the sense that they demand both storage and processing operations at encoding.

- *Listening Span Task* [43, 44] consists of simple three-word sentences presented in groups of three-to-six sentences. The task is to answer 'yes' or 'no' depending on whether the sentence makes sense or is nonsense. Then the subject is

required to recall the first or last word in the set of sentences without knowing which in advance.

- *The Picture Span* [45] contains pictures of faces that are presented one at a time for 1 second in a central position on the computer screen. The subject has to answer whether it is a female or a male face. When the last picture (face) has faded, all pictures in the current span are presented randomly. The task is then to recall in which order the faces were presented initially. Span length is increased by one picture at a time if the subject successfully recalls at least one of the three spans of a certain length. Span lengths are two-to-seven pictures. The tests were administered in the same order for each person and on each test occasion.

To contrast the putative training effect on the complex WM span tasks with other kinds of WM tasks, a simple WM storage task, Digit Span, was also included with a forward recall instruction [8].

WM related activities, quality-of-life and health. All subjects were interviewed by an occupational therapist using the *Canadian Occupational Performance Measure-COPM* [46]. The COPM is a client-centred outcome measure focusing on the interdependence between person, environment and occupation. When using the COPM as an outcome measure, the clients evaluate their occupational performance and their satisfaction with performance on the basis of their defined problems in areas of self-care, productivity and leisure. A semi-structured interview is conducted to encourage the patient to identify and discuss specific activities that give him or her difficulty. Clients are always prompted to identify activities that they want to do, need to do or are expected to do.

For this study the current subjects were asked to identify WM-related occupational and participating problems and rate 'performance' and 'satisfaction with performance' before and at 20 weeks after the training program. Ratings for 'performance' and 'satisfaction with performance' for each identified problem were done on two 1–10 point scales; 1 = not able to do it/not satisfied at all, 10 = able to do it extremely well/extremely satisfied. The subjects were not allowed to look at the first scoring when they did their final scoring 25 weeks later. They also completed the *EQ-5D questionnaire* about their health-related quality-of-life and overall health [47, 48] before and 20 weeks after training.

WM training. The individuals performed their WM training program, QM, on a personal computer (PC), 45–60 minutes of intense training per day, 5 days per week, for 5 weeks. All individuals trained

in pairs in a separate quiet room at the Department of Rehabilitation. One of three certified coaches was in the room all the time and special feedback was provided once a week beside the continuous statistics, which the patients could follow themselves on the computer.

The training program specifies different visuo-spatial and verbal WM tasks. For example, the visuo-spatial WM tasks require the subject to remember the position of stimuli in a four-by-four grid and then they are asked to reproduce stimuli in the same order, in the reverse order or in the grid after the grid has been rotated. The verbal WM tasks require the subject to remember sequences of letters and digits forwards and/or backwards. Each task is introduced by a speaker voice and the person responds by localizing and remembering the stimuli. The subjects respond using a computer mouse (for a more detailed description see [29, 34, 35]). The difficulty level of each training task was automatically adapted to each individual's WM capacity. Thus, WM load was increased according to each individual's performance levels. An individual's WM results indicate where he/she had managed all trials at a certain level.

After finishing the training program, each individual received an Improvement index, which is a measure of his/her improvement during the training period, calculated by subtracting the Start index from the Max index. The Improvement index is based on the person's results from three standard selected exercises, which are one grid, one letter span and one digit span task, and on the assumption that the mean daily performance on these three tasks are *t*-standardized ($M = 50$, $SD = 10$) and aggregated into a composite score, which is the Improvement index. The Start index is calculated using the results from day 2 and 3, while the Max index is calculated using the results from the two days with the best performances during the training period.

After completion of the 20-week follow-up, each group was offered a debriefing where the whole group received feedback on their results. Individual results as well as group results were analysed and each individual was then encouraged to give feedback on their experiences. They were also asked to provide specific feedback regarding any long-term effects from the training. The individuals were also encouraged to give their view on how the training was organized.

Design

The study is a controlled study with a cross-over design. To control for test-re-test effects and to ensure the outcome is truly from the training and not due to spontaneous cognitive improvement, the individuals were randomized into two groups at the

beginning of the study, through 'drawing of lots'. Ten individuals were included in group I and 11 individuals in group II. There was no age difference ($p = 0.973$) or difference in duration since the onset of the injury/illness ($p = 0.912$) between the two groups. First, Group I received systematic WM training for 5 weeks, whereas Group II did not receive any training during the same period. Four weeks after the Group I participants had had their training, WM functions were assessed in both groups. Thus, Group II performed the neuropsychological test battery twice without any training in between to control for non-specific test-re-test effects. Then, Group II received the same training for 5 weeks. For all subjects, a short-term follow-up was carried out 4 weeks after training and then a long-term follow-up was performed 20 weeks after training for all subjects. A flow-chart of the design and test occasions can be seen in Figure 1.

The study was approved by the ethical committee at the University of Linköping. Written informed consent was obtained from the patients before participation in accordance with the Declaration of Helsinki.

Statistical methods

Descriptive analysis, qualitative analysis and quantitative analysis have been used. For quantitative analysis, the Student *t*-test, the Mann Whitney's U-test, the Wilcoxon signed-rank test and Spearman's rank correlation coefficient were all used. When comparing pairs of tasks (the non-complex Digit Span forwards results with the results of the complex WM results), Paired samples *t*-tests were used (one-tailed). An overall significance level of 0.05 was used. Considering the repeated analyses of paired data when looking at the outcome from neuropsychological tests, the Bonferroni correction was used to set the statistical significance level to 0.008 ($\alpha = 0.05/6 = 0.008$). All statistical analyses were made using Statistica 9.0 and SPSS for Windows version 17.

Results

All individuals completed the WM training programme. The number of training days varied between 21–25 days ($M = 23.2$; $SD = 1.6$).

Questionnaire

According to the questionnaire given before training, the most common self-reported difficulties in daily living situations were remembering long instructions (86%) and solving problems (86%). More than 80% reported that they were absent-minded and more

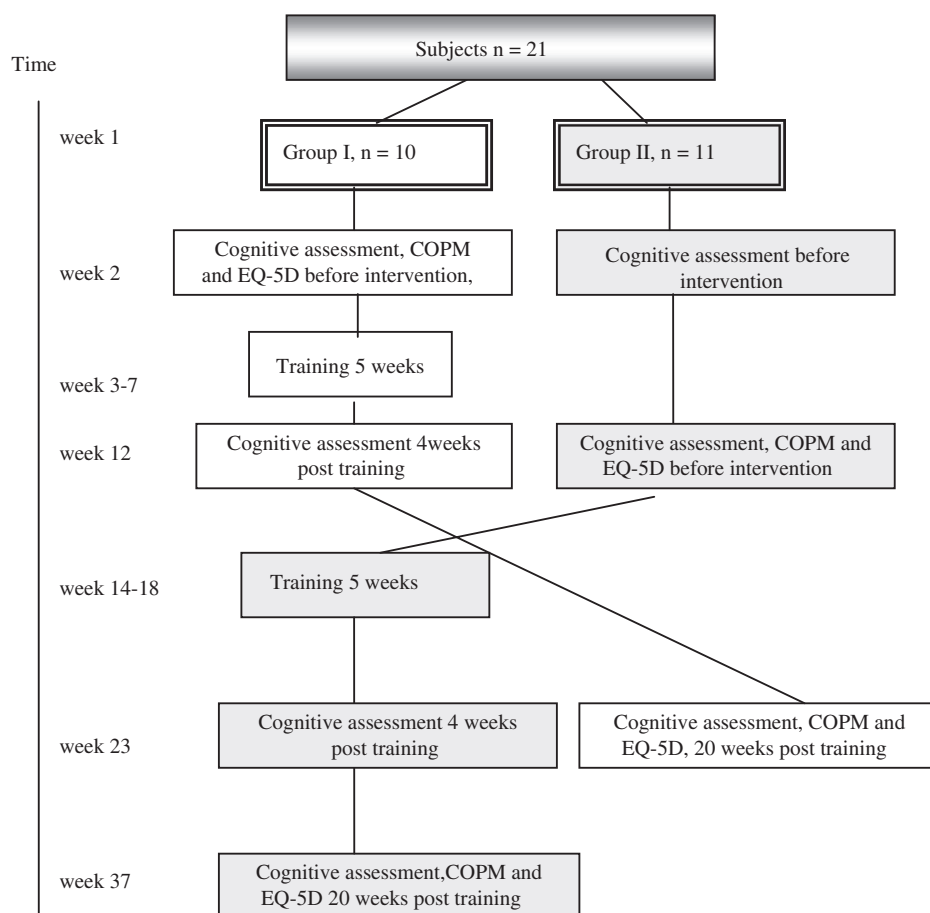


Figure 1. Flow-chart of the study design and assessment occasions.

Table IV. Neuropsychological test results for Group I and II, baseline pre-training.

Test	Group I <i>M</i> (SD) (<i>n</i> = 10)	Group II <i>M</i> (SD) (<i>n</i> = 11)	Mann-Whitney's U-test, exact <i>p</i> -value
PASAT (max 60 digits)	39.1 (11.9)	36.1 (11.7)	0.70
Listening Span Test (max 54 words)	22.6 (3.5)	21.5 (6.7)	0.56
Picture Span (max 17 spans)	8.8 (2.9)	6.9 (2.4)	0.17
Block Span forwards (max 14 spans)	7 (1.6)	7.3 (2.2)	0.86
Block Span backwards (max 14 spans)	6.9 (1.6)	6.7 (2.3)	0.97
CWIT inhibition/switching (seconds)	64.5 (20.2)	86.6 (37)	0.13

than 75% expressed that they often make mistakes when they are in a hurry.

There was no difference between the two groups in the neuropsychological WM test results at baseline before the training (Table IV). Table V shows that there was no difference in test results between first and second test situation for Group II, indicating that there was no training effect from re-testing or change over time due to other factors.

WM tests

As seen in Table VI, there was a significant difference in the non-trained WM tests, PASAT,

Listening Span, Block Span backwards and CWIT inhibition/switching between the time the baseline was established and 4 weeks after training as well as 20 weeks after training. For Picture Span, there was a significant difference 4 weeks after training, whereas the difference was not significant 20 weeks after training ($p = 0.012$). For Block Span forwards, the difference 4 weeks after training was nearly significant, while it was significant 20 weeks after training compared to the baseline. One person did not show up at the follow-up 20 weeks after training.

To compare the training effect on the *complex* WM span tasks with a *simple* WM storage task (the Digit Span task with a forward recall instruction) a *t*-test

Table V. Neuropsychological test results for Group II controlling for test-re-test effects.

Test	Group II ($n = 11$)		
	Baseline M (SD)	Four weeks after Group I training	Wilcoxon signed-rank test, exact p -value
PASAT (max 60 digits)	36.1 (11.7)	36.3 (14.0)	0.58
Listening Span Test (max 54 words)	21.5 (6.7)	23.3 (7.6)	0.17
Picture Span (max 17 spans)	6.9 (2.4)	7.9 (1.2)	0.13
Block Span forwards (max 14 spans)	7.3 (2.2)	7.3 (1.3)	0.40
Block Span backwards (max 14 spans)	6.7 (2.3)	7.6 (1.4)	0.26
CWIT inhibition/switching (seconds)	86.6 (37)	88.5 (38.0)	0.62

Table VI. Neuropsychological test results at baseline, 4 and 20 weeks after training.

Test	Baseline M (SD) ($n = 21$)	Four weeks after training M (SD) ($n = 21$)	Wilcoxon signed-rank test, p -value	Twenty weeks after training M (SD) ($n = 20$)	Wilcoxon signed-rank test, p -value
PASAT (max 60 digits)	37.5 (11.1)	44.5 (11.8)	<0.001	44.55 (12.11)	<0.001
Listening Span Test (max 54 words)	22 (5.2)	26.1 (5.1)	<0.001	26.85 (5.27)	<0.001
Picture Span (max 17 spans)	7.8 (2.7)	9.8 (2.5)	0.002	9.7 (2.87)	0.012 (ns)
Block Span forwards (max 14 spans)	7.1 (1.9)	8.5 (1.4)	0.009 (ns)	8.7 (1.69)	0.002
Block Span backwards (max 14 spans)	6.8 (1.9)	8.2 (1.6)	0.005	8.75 (1.55)	0.001
CWIT inhibition/switching (seconds)	76.1 (30.8)	63.7 (25.2)	<0.001	64.55 (26.34)	0.002

Paired samples (one-tailed) showed that the relative remaining long-term WM training effect (20 weeks post-training) was significantly higher for PASAT compared to Digit Span forwards, $t(19) = 1.87$, $p < 0.05$, and for Listening Span compared to Digit Span forwards, $t(19) = 1.88$, $p < 0.05$. At the baseline, the performance level of Digit Span forwards was $M = 8.3$ (SD = 1.4) ($n = 21$) and 20 weeks after training the performance level was $M = 8.9$ (SD = 0.89) ($n = 20$).

WM training

All 21 individuals increased their WM index when comparing the Start index (baseline) and the Max index. The Start index was $M = 73.7$ (SD = 9.7) and the Max index was $M = 93.4$ (SD = 13.7). The difference varied between 9.0–35.0. Table II shows the individuals' WM index results.

There was a significant overall improvement when subtracting the Start index from the Max index for WM-trained tasks ($p < 0.001$) as tested by the Wilcoxon signed-rank test. In addition, the difference between the Start index values and the Max index values was found to be significant by means of a t -test for dependent samples, $t = 13.55$, $p < 0.001$. The effect size was large, $r = 0.95$, as calculated by the formula $r = \sqrt{\frac{t^2}{t^2 + df}}$ [49].

Although there was no pattern between improvement results and diagnosis, the lowest Improvement

index belonged to a person with a linguistic slowness (Table II). Nor could one find any correlation between improvement and age ($r_s = -0.16$). The time for peak Max index differed between the subjects.

However, the best results were typically obtained during the second part of the training period. For 74% of the subjects their peak index was obtained during the last 30% of the training period. Nine per cent of the subjects reached their peak index already during the first 50% of the training period.

WM-related activities, quality-of-life and health

Figure 2 shows the categorized individually prioritized problems with occupational performance, defined before the training started. The three most frequently defined occupational performance problems were identified in the following areas: Community management, Household management and Socialization. Community management included activities such as communicating with public authorities and handling one's finances. Household management included activities such as shopping and cooking. Socialization included a conversation with several persons.

Results from the COPM interviews show a difference in estimated performance of prioritized occupations, before training vs 20 weeks after training, $p < 0.05$. An even bigger difference was found in

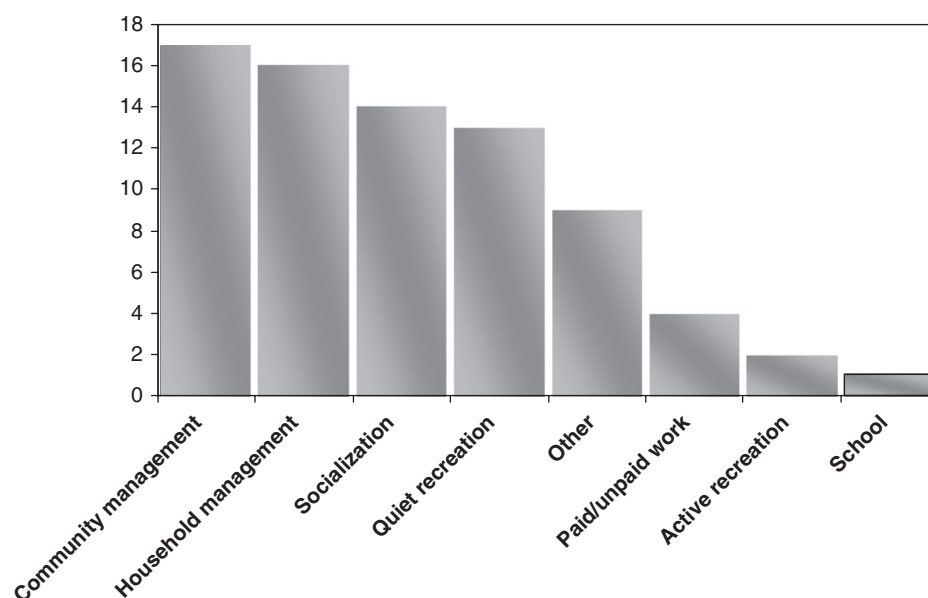


Figure 2. The individually identified problems with occupational performance related to WM were sorted into eight occupational areas. Each column represents the number of activities identified within that area.

Table VII. Canadian Occupational Performance Measures and EQ-5D at baseline and 20 weeks after training, $n = 20$.

	Baseline <i>M</i> (SD)	Twenty weeks after training <i>M</i> (SD)	Wilcoxon signed-rank test, <i>p</i> -value
COPM Performance	4.4 (1.5)	5.4 (1.9)	0.016
COPM Satisfaction	3.7 (2.1)	5.3 (2.1)	<0.001
EQ-5D	0.7 (0.3)	0.8 (0.2)	0.209 (ns)
Health self-rating (VAS)	58.7 (22.7)	67.7 (18.4)	0.020

estimated satisfaction with performance before vs 20 weeks after training, $p < 0.001$ (Table VII). There was no difference ($p > 0.05$) in health-related quality-of-life, as measured by the EQ-5D questionnaire, while there was a significant difference in the health self-rating VAS ($p < 0.05$).

Discussion

This study evaluates short- and long-term effects of WM training for a group of patients suffering from WM impairments after acquired brain injury. In addition, the study evaluates whether the training had an impact on the patients' occupational performance, satisfaction with performance and their overall health. All subjects had WM impairments, which was the main inclusion criterion, defined by the participants' WM index on WAIS III compared to other WAIS indices. In addition, they experienced WM problems in daily living. Because this study focused on function and activity [50] rather than on

diagnosis, it looked at individuals with different diagnoses.

Trained WM tasks

There was a significant improvement in the WM-trained tasks as measured by the difference between the Start index at baseline and the Max index. All 21 individuals increased their index. The effect size was large. The individual improvements differed between 9.0–35.0, which suggests a large variation between the subjects. One reason is that the individuals started at different index levels (Table II). Another reason might be that the study group was small and heterogeneous. Given a larger group, diagnosis might have shown a difference in the ability to benefit from this cognitive rehabilitation programme due to specific neuropsychological deficits. A third reason might be that some individuals have experienced a lack of social support and high levels of stress and demands in their private life, time consuming travelling and the like. Motivation for training might also have had an impact on the results. However, the results replicate earlier studies [29, 30, 35].

Non-trained WM tasks

The WM training had an effect on the non-trained neuropsychological WM test results as well. This is interesting from a transfer-of-training point of view. The neuropsychological examination contained verbal as well as visual-spatial WM tests with more or less WM loading due to WM task complexity. There was a significant difference in the answers

4 weeks after training in the WM demanding tests PASAT, Listening Span, Block Span backwards, CWIT inhibition/switching and Picture Span ($p < 0.01$). With respect to PASAT and Block Span, the results replicate previous studies [29, 30, 35]. PASAT has been one of the most sensitive tests for measuring improvement after attention- and WM-related training [24, 51, 52]. In this study, the improvement of test results lasted at least 20 weeks after training compared to baseline ($p < 0.01$) except for Picture Span, which was only borderline significant ($p = 0.012$) 20 weeks after training.

Related to the sensitivity of tests, this study focused on complex WM tasks to evaluate the generalization of the WM training effect. PASAT, Listening Span, Picture Span and CWIT inhibition/switching require combined WM functions of temporary storage, processing, as well as executive attention. Because of the composite nature of these tests, they are probably more related to complex communicative WM tasks in daily living and working activities than, for instance, Block Span forwards and Digit Span forwards. The fact that the transfer effect was more pronounced for PASAT and Listening Span compared to Digit Span forwards is important because the current WM training may show important transfer to daily life beside the high sensitivity to change [53]. Thus, it is important to understand the cognitive processes targeted by the training programme for understanding and evaluating why transfer to other non-trained tasks occur [54].

One question is whether improvements may reflect test-re-test effects. The authors made a controlled study with a cross-over design to control for test-re-test effects and to ensure that the outcome was truly from the WM training and not due to spontaneous cognitive improvement. The individuals were therefore randomized into two groups. Group II, the 'control group', did not receive any training during the period when Group I received the WM training, and performed the neuropsychological test battery twice without any training in between. No difference was found in test results between first and second test situation for Group II, indicating that there was no test-re-test effect or change over time due to for instance spontaneous change.

The relatively larger effects of the current WM training on the complex WM tests Listening Span and PASAT compared to the simple WM storage Digit Span forwards indicate substantial transfer effects of training in most cases and that test-re-test effects are probably minor relative to the effects due to the actual WM training. In addition, the WM training effect size was *large*. This is an important aspect of the data since it is claimed that transfer was specifically related to WM training.

WM-related activities

Another purpose of the study was to evaluate whether a training effect could be transferred to WM-related activities and patients' evaluation of their quality-of-life and health. Transfer to the subjects' self-reported occupational performance was found. There were significant improvements in COPM self-estimation of occupational performance ($p < 0.05$) as well as for satisfaction with performance ($p < 0.01$), findings that suggest a training effect on cognitive functioning in daily living. The results agree with some other studies that show transfer from training on demanding WM tasks to cognitive functions such as attention, impulsivity control, problem-solving and to cognitive functioning in daily living [29, 30, 33, 35, 55].

At the meeting offered to all subjects after the 20-week follow-up everyone was encouraged to give feedback on their experiences and long-term effects from the training. The subjects mentioned that after training they felt more alert and that they could resist distraction from irrelevant stimuli better in situations without noise. For example, this improvement was noticeable when helping children doing their homework, remembering codes and phone-numbers, following a road-description and shopping. However, in stimuli-intense situations they still experienced problems with concentration.

Clinical implications

This study shows that an individually structured and intense WM training can improve a person's WM function. It found effects at the function level of non-trained WM tasks as well as at the activity level [45] of self-reported performance and satisfaction with performance and for global health ratings.

The training is intense, demanding and tiring. Therefore, the authors recommend offering it to individuals with a fairly well preserved cognitive function, as was the case for the subjects in the current study, and for persons with a strong motivation for training. The training can be given in the post-acute phase in clinical out-patient services. The motivation varied among the present individuals and probably persons who are highly motivated and who are planning to return to work will profit most from the training. It also prepares the subjects for returning to work since it also provides a daily structure.

Training 5 days per week for 5 weeks was not an optimal plan for all individuals. A more flexible structure, where the individuals could come 4 or 5 days per week, would have suited everyone. However, the consequence of this is a prolonged training period, from 5 to maybe 7 weeks. The authors also suggest that the training takes place at the clinic—at least at the beginning. This was

something that the individuals agreed on, re-enforcing the importance of coaching, meeting with others and training in a peaceful and quiet environment. Some individuals suffering from minor aphasic impairment had problems with some of the exercises. It would have been helpful if one or two of the most linguistically demanding exercises could have been excluded.

Study limitation

A limitation of the study is the small number of subjects with different diagnoses. Strengths are the long-term follow-up at 5 months, the computerized training program was evaluated on adult persons by highly demanding neuropsychological WM tests and all participants improved despite having different initial diagnoses. Despite the demanding program, there were no drop outs from the training and just one drop out from the 20 weeks follow-up.

The conclusion is that WM training with this computerized method can improve cognitive functioning in the current sample of patients with acquired brain injury, measured by neuropsychological complex WM tests and by the individuals' ratings of occupational problems and overall health. The training programme might also more generally improve WM for persons with cognitive impairments after brain injury and it will probably have an impact on rehabilitation, returning to work and on daily activities. The training programme could also have an impact on the individual's wellbeing.

The purpose of clinical research is to enhance the knowledge base related to practice. This study reveals that a computerized WM training programme can be effective and a resource that should be offered by the department. A continuous long-term follow-up of occupational performance and daily living for an increased number of subjects should be an interesting and a valuable addition to this study. The authors propose further research with a larger study group, including sub-groups with different diagnoses, to confirm the current findings and select for whom this cognitive rehabilitation programme is most suitable.

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References

1. Baddeley AD. Human memory: Theory and practice. Hove, UK: Laurence Erlbaum Associates, Inc; 1990.
2. Goldman-Rakic P. Circuitry of primate prefrontal cortex and regulation of behavior by representational memory. In: Plum F, editor. Handbook of physiology, the nervous system, higher functions of the brain. Volume 5. Bethesda, MD, American Physiological Society. 1987. p 373–417.
3. Kolb B, Wishaw IQ. Fundamentals of human neuropsychology. New York: Worth Publishers; 2003.
4. Nyberg L. Kognitiv neurovetenskap. Lund: Studentlitteratur; 2002.
5. Wager T, Smith D. Neuroimaging studies of working memory: A metaanalysis. *Cognitive, Affective and Behavioral Neuroscience* 2003;3:255–274.
6. Rönnerberg J. Cognition in the hearing impaired and deaf as a bridge between signal and dialogue: A framework and a model. *International Journal of Audiology* 2003;42:68–76.
7. Rönnerberg J, Rudner M, Foo C, Lunner T. Cognition counts: A working memory system for ease of language understanding (ELU). *International Journal of Audiology* 2008;47:171–177.
8. Wechsler D. Wechsler adult intelligence scale. 3rd ed. San Antonio, TX: The Psychological Corporation; 1997.
9. Klingberg T, Roland PE. Right prefrontal activation during encoding, but not during retrieval, in a non-verbal paired associates task. *Cerebral Cortex* 1998;8:73–79.
10. Westerberg H, Klingberg T. Changes in cortical activity after training of working memory – a single-subject analysis. *Physiology and Behavior* 2007;92:186–192.
11. Olesen PJ, Westerberg H, Klingberg T. Increased prefrontal and parietal activity after training of working memory. *Nature Neuroscience* 2004;7:75–79.
12. McNab F, Varrone A, Farde L, Jucaite A, Bystritsky P, Forssberg H, Klingberg T. Changes in cortical dopamine D1 receptor binding associated with cognitive training. *Science* 2009;323:800–802.
13. D'Esposito M, Postle B, Rypma B. Prefrontal cortical contributions to working memory: Evidence from event-related fMRI studies. *NeuroImage* 2000;8:176–187.
14. Duncan J, Owen AM. Common regions of the human frontal lobe recruited by diverse cognitive demands. *Trends in Neurosciences* 2000;23:475–483.
15. Sylvester C, Wager T, Lacey S, Hernandez L, Nichols T, Smith E, Jonides J. Switching attention and resolving interference: fMRI measures of executive functions. *Neuropsychologia* 2003;41:357–370.
16. Brooks N, Mc Kinlay A, Symington C, Beattie A, Campsie L. Return to work within the first seven years of severe head injury. *Brain Injury* 1987;1:5–19.
17. Cicerone K. Remediation of 'working attention' in mild traumatic brain injury. *Brain Injury* 2002;16:185–195.
18. Lezak M. Neuropsychological assessment. New York: Oxford University Press; 2004.
19. Robertson I, Murre J. Rehabilitation of brain damage: Brain plasticity and principles of guided recovery. *Psychological Bulletin* 1999;125:544–575.
20. Cicerone K, Dahlberg C, Kalmar K, Langenbahn DM, Malec JF, Bergquist TF, Felicetti T, Giacino JT, Harley JP, Harrington DE, et al. Evidence-based cognitive rehabilitation: Recommendations for clinical practice. *Archives of Physical Medicine and Rehabilitation* 2000;81:1596–1615.
21. Cicerone K, Dahlberg C, Malec JF, Langenbahn DM, Felicetti T, Kneipp S, Ellmo W, Kalmar K, Giacino JT, Harley JP, Laatsch L, Morse P, Catahese J. Evidence-based cognitive rehabilitation: Updated review of the literature from (1998) through (2002). *Archives of Physical Medicine and Rehabilitation* 2005;86:1681–1692.

22. Turner-Stokes L, Nair A, Sedki I, Disler PB, Wade DT. Multi-disciplinary rehabilitation for acquired brain injury in adults of working age. *Cochrane Database of Systematic Reviews* 2005, Issue 3. Art No. CD004170. DOI:10.1002/14651858.CD004170.pub2.
23. Turner-Stokes L. Evidence for the effectiveness of multi-disciplinary rehabilitation following acquired brain injury: A synthesis of two systematic approaches. *Journal of Rehabilitation Medicine* 2008;40:691–701.
24. Flavia M, Stampatori C, Zanotti D, Parrinello G, Capra R. Efficacy and specificity of intensive cognitive rehabilitation of attention and executive functions in multiple sclerosis. *Journal of Neurological Sciences* 2010;15:101–105.
25. Parente R, DiCesare A. Retraining memory: Theory, evaluation, and applications. In: Kreutzer J, Wehman P, editors. *Cognitive rehabilitation for persons with traumatic brain injury: A functional approach*. Baltimore, MD: Paul H. Brookes Publishing; 1991. pp 147–163.
26. Raskin SA. Neuropsychological management of mild traumatic brain injury. In: Raskin S, Mateer C, editors. *New York: Oxford University Press*; 2000. pp 93–112.
27. Tam SF, Man WK. Evaluating computer-assisted memory retraining programmes for people with post-head injury amnesia. *Brain Injury* 2004;18:461–470.
28. Craik FI, Winocur G, Palmer H, Binns MA, Edwards M, Bridges K, Glazer P, Chavannes R, Stuss DT. Cognitive rehabilitation in the elderly: Effects on memory. *Journal of the International Neuropsychological Society* 2007;13: 132–142.
29. Klingberg T, Forssberg H, Westerberg H. Training of working memory in children with ADHD. *Journal of Clinical and Experimental Neuropsychology* 2002;24: 781–791.
30. Klingberg T, Fernell E, Olesen P, Johnson M, Gustavsson P, Dahlström K, Gillberg CG, Forssberg H, Westerberg K. Computerized training of working memory in children with attention-deficit/hyperactivity disorder—a controlled, randomized, double-blind, trial. *Journal of the American Academy of Child and Adolescent Psychiatry* 2005;44: 177–186.
31. Westerberg H, Hirvikoski T, Forssberg H, Klingberg T. Visuo-spatial working memory span: A sensitive measure of cognitive deficits in children with ADHD. *Child Neuropsychology* 2004;10:155–161.
32. Westerberg H, Klingberg T. Changes in cortical activity after training of working memory – a single-subject analysis. *Physiology and Behavior* 2007;92:186–192.
33. Holmes J, Dunning DL, Gathercole S. Adaptive training leads to sustained enhancement of poor working memory in children. *Developmental Science* 2009;12:9–15.
34. Westerberg H. Working memory: Development, disorders and training. Doctoral dissertation. Karolinska institute, Stockholm, Sweden; 2004.
35. Westerberg H, Jacobaeus H, Hirvikoski T, Clevberger P, Östensson ML, Bartfai A, Klingberg T. Computerized working memory training after stroke – A pilot study. *Brain Injury* 2007;21:21–29.
36. Diagnostic and statistical manual of mental disorders. 4th Ed. DSM-IV, Washington, DC: American Psychiatric Association; 1994.
37. Cogmed. 2007. Available online at: http://download.cogmed.com/research/indications_training_071113.doc, accessed.
38. Gronwall D. Paced auditory serial-addition task: A measure of recovery from concussion. *Perceptual and Motor Skills* 1977;44:367–373.
39. Delis D, Kaplan E, Kramer H. D-KEFS. Technical Manual. San Antonio, TX. The Psychological Corporation; 2001.
40. Stroop J. Studies of interference in serial verbal reactions. *Journal of Experimental Psychology* 1935;18:643–662.
41. Kaplan E, Fein D, Morris R, Delis D. WAIS-R as a neuropsychological instrument. New York: The Psychological Corporation; 1981.
42. Nyman H, Johansson C. WAIS-III NI: WAIS III som neuropsykologiskt instrument. Stockholm: Hartcourt Assessment; 2004.
43. Daneman M, Carpenter P. Individual differences in working memory and reading. *Journal of Verbal Learning and Verbal behaviour* 1980;19:450–466.
44. Rönnerberg J, Arlinger S, Lyxell B, Kinnefors C. Visual evoked potentials: Relation to adult speechreading and cognitive function. *Journal of Speech and Hearing Research* 1989;32: 725–735.
45. Leven A, Lyxell B, Andersson J, Danielsson H, Rönnerberg J. Prospective memory, working memory, retrospective memory and self-rated memory performance in persons with intellectual disability. *Scandinavian Journal of Disability Research* 2008;10:147–165.
46. Law M, Baptiste S, McColl MA, Carswell-Opzoomer A, Polatajko H, Pollock N. *Canadian Occupational Performance Measure Manual*. Toronto, ON: CAOT Publications; 1991.
47. EuroQol. A new facility for the measurement of health-related quality of life. *Health Policy* 1990;16:199–208.
48. Brooks R. EuroQol: The current state of play. *Health Policy* 1996;37:53–72.
49. Field A. *Discovering statistics using SPSS*. 3rd ed. CA: Sage Publications; 2009.
50. World Health Organization. *International classification of functioning, disability and health: ICF*. Geneva: WHO; 2001.
51. Gray JM, Robertson I, Pentland B, Anderson S. Microcomputer-based attentional retraining after brain damage: A randomized group controlled trial. *Neuropsychological Rehabilitation* 1992;2:97–115.
52. Niemann H, Ruff RM, Baser CA. Computer assisted attention retraining in head injured individuals: A controlled efficacy study of an out-patient program. *Journal of Consulting and Clinical Psychology* 1990;58:811–817.
53. Hannon B, Daneman M. A new tool for measuring and understanding individual differences in the component processes of reading comprehension. *Journal of Educational Psychology* 2001;93:103–128.
54. Dahlin E, Stigsdotter Neely A, Larsson A, Bäckman L, Nyberg L. Transfer of learning after updating mediated by the striatum. *Science* 2008;13:1155–1512.
55. Jaeggi SM, Buschkuhl M, Jonides J, Perrig WJ. Improving fluid intelligence with training on working memory. *Proceeding of the National Academy of Sciences of the United States of America* 2008;105:6829–6833.